



NITI Aayog



SECTOR AT A GLANCE 01

WASH Sector in India

An Overview

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WASH SECTOR IN INDIA

AN OVERVIEW

The '**Sector-at-a-Glance**' provides an easy-to-read overview and basic understanding of a sector, sub-sector, or thematic area. The facts and insights on the sector are supported by relevant data and best practices from credible government and multilateral sources. The document covers the structure, key trends, opportunities and challenges, government initiatives, and the way forward for the sector.

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Overview

Water, sanitation and hygiene (WASH) provision and uptake are vital enablers for human health, wellbeing and dignity, while also unlocking broader socio-economic gains. Over the last decade, India has made remarkable strides in expanding access to safe drinking water, toilets and hygiene facilities for its citizens. This has been made possible by the country's marquee schemes such as Swachh Bharat Mission (SBM), Jal Jeevan Mission (JJM), Atal Mission for Rejuvenation and Urban Transformation (AMRUT), alongside other related initiatives under the National Rural Health Mission, MGNREGA and various State programs.

Coverage for water and sanitation services has improved significantly. 97.1% of people in urban areas and 78.7% of people in rural areas have access to improved latrines. 97% of persons in urban areas and 95% of people in rural areas have access to improved drinking water. These significant achievements are a result of focused mission mode approach adopted by the government in the last ten years.

However, despite the significant progress, specific gaps do remain that need sustained attention going forward. 15% of households across the country do not have access to toilets and continue to defecate in the open. In urban India, 39% of households do not have access to piped water supply while 3% lack toilets, particularly impacting slum settlements and floating migrant populations. Menstrual health and hygiene continue to be a challenge for large segments of rural and urban women.

While platforms for services are now largely in place after the infrastructural push of the last few years, there is an urgent need to shift focus to the sustainability of outcomes and assets created. A range of issues will need to be addressed - intermittent and inadequate quantities of piped water supply to households, lack of maintenance of village taps and handpumps, poor adherence to waste segregation practices, lack of used water treatment capacity and efficiency, gaps in safe reuse of treated wastewater, continued reliance on unscientific sewage management, and persistence of taboos restricting toilet usage and adoption of hygienic menstrual practices. The human and systemic capabilities now need enhancement and community institutions require support to ensure upkeep of infrastructure and drive habit formation at scale. Similarly, new operating models for waste management and more so for faecal management – such as STPs, FSTPs – and also for safe water delivery will need significant investment, innovation, and support for them to become scalable and self-sustaining.

The newly launched Phase 2 programs of SBM-G and AMRUT already align to this roadmap with an emphasis on waste management, behaviour change related to waste management, toilet usage, and safe drinking water, as well as the strengthening of rural and urban institutions. However, much work lies ahead to universalize services, address persistent social taboos around sanitation practices, build climate resilience in WASH infrastructure, leverage emerging digital tools for real-time monitoring and responsible pricing of WASH services. Finally, nurturing collaborations between communities, civil society actors, technical experts and different tiers of government will be the key to enabling evidence-based planning and accountable implementation.

India now stands at an important juncture with respect to its WASH trajectory. Having made tremendous strides in expanding coverage, it must now consolidate the gains made and drive sustainable impact.

This note on WASH highlights key statistics, key challenges and opportunities, and also some ideas for the way forward.

Figure 1: Key WASH statistics

Toilets

97.1%

Of households in urban areas had access to an improved toilet facility as of 2020-21

78.7%

Of households in rural areas had access to an improved latrine as of 2020-21

Access to water

97.2%

Of persons in urban areas had used an improved source of drinking water as of 2020-21

95%

Of persons in rural areas had access to an improved source of drinking water as of 2020-21

Waste management

86%

Of ODF cities (4,576) were designated as ODF Plus (3,913) (i.e., cities that sustain ODF status and ensures functional and well-maintained community and public toilets) as of March 2024

90%

Of villages in India were designated ODF Plus (i.e., villages that sustain ODF status and have solid or liquid waste management arrangements) as of March 2024

Menstrual hygiene

89.4%

Share of urban women (15-24 years) that used hygienic methods of protection during their menstrual period in 2019-21

72.3%

Share of rural women (15-24 years) that used hygienic methods of protection during their menstrual period in 2019-21

Handwashing

92.7%

Persons in urban areas with access to a hand washing facility with water and soap or detergent within the premises

77.4%

Persons in rural areas with access to a hand washing facility with water and soap or detergent within the premises

Source: MoSPI, Multiple Indicator Survey of India 2020-21, 2023; Ministry of Health and Family Welfare, National Family Health Survey (NFHS-5) 2019-21, 2021; Ministry of Jal Shakti, SBM-G Dashboard, accessed March 2024; Ministry of Housing and Urban Affairs, SBM-Urban 2.0, accessed March 2024.

Note: Improved latrines include flush/pour-flush to piped sewer system, flush/pour-flush to septic tank, flush/pour-flush to twin leach pits/single pit, ventilated improved pit latrine, pit latrine with slab and composting latrine. Improved sources of drinking water include bottled water, piped water into dwelling, piped water to yard/plot, piped water from neighbour, public tap/standpipe, tube well, hand pump, protected well, public tanker truck, private tanker truck, protected spring, and rainwater collection.

(1), (2), (3), (4) NITI Aayog, [Composite Water Resources Management: Performance of States](#), 2018

Sub-sectoral trends: Urban WASH

This section presents a summary view of relevant statistics and trends related to the following sub-sectors within the urban WASH sector:

- Urban waste management
- Urban household water supply
- Urban toilet and waste infrastructure
- Urban handwashing with soap
- Urban menstrual hygiene management

Urban Waste Management

Urban waste management in India focuses on the organized collection, transportation, and disposal of municipal solid waste in urban areas. The process includes waste segregation, transportation to treatment plants, and initiatives for recycling and waste-to-energy, aimed at minimizing environmental impact and promoting responsible waste disposal practices.

Key urban waste management statistics

70%

India's solid waste processing capacity

60%

Share of toilets in urban India relying on "on-site" sanitation systems

72,368 MLD*

Sewage generation in urban areas (higher than treatment capacity)

Source: Waste processing capacity- NITI Aayog, Waste Wise Cities- Best practices in municipal solid waste management, 2021; Toilets relying on OSS- NITI Aayog, Faecal sludge and septage management in urban areas, 2021; Sewage generation- NITI Aayog, Urban Wastewater Scenario in India, 2022

Note: * MLD: Million litres per day

India's solid waste management approach has undergone a significant shift over the past decade: The last few years have witnessed a growing focus on waste segregation and door-to-door collection in India and a significant increase in the country's municipal solid waste processing capacity from 18% in 2014 to 70% in 2021.¹ Guidelines for the use of waste, such as use of plastic waste in cement kilns, have also been issued.²

Urban Household Water Supply

Urban household water supply involves the establishment of water supply infrastructure, treatment plants, and distribution networks to ensure a consistent and potable water supply for households in urban areas. The goal is to meet the increasing demand for water in rapidly growing urban centres, ensuring access to clean water and promoting public health. AMRUT 2.0 run by MoHUA focuses on enhanced access to universal water supply through functional taps to all urban households.

Key urban household water supply statistics

INR 2,99,000 crore

Indicative financial outlay for AMRUT 2.0 scheme for urban household water supply

134 lakh

Urban water tap connections provided through AMRUT 2.0

¹ NITI Aayog, Waste Wise Cities, 2021

² CPCB, Guidelines for Co-processing of Plastic Waste in Cement Kilns, 2017

97.2%

Urban residents have access to improved sources of drinking water

61.5%

Urban residents having access to piped water into dwelling/yard/plot

Source: Ministry of Housing and Urban Affairs, AMRUT 2.0 Press Release, 2022; Ministry of Housing and Urban Affairs, AMRUT Scheme, 2022; MoSPI, Multiple Indicator Survey in India, 78th Round, (2020-21), 2023

Note: * Improved sources include (i) bottled water, (ii) piped water into dwelling, (iii) piped water to yard/plot, (iv) piped water from neighbour, (v) public tap/standpipe, (vi) tube well, (vii) hand pump, (viii) protected well, (ix) public tanker truck, (x) private tanker truck, (xi) protected spring and (xii) rainwater collection

Access to principal source of drinking water for exclusive use of household has also increased:

The number of urban households who reported access to a principal source of drinking water that was for the exclusive use of households stood at ~65% in 2021, a marked improvement from ~47% in 2012.³

More individuals in urban areas now have access to sufficient⁴ water:

76% of urban residents reported having access to an improved source of drinking water located within household premises that was sufficient for the household in 2021, compared to 72% in 2018.⁵

Urban Toilet And Waste Infrastructure

Urban toilet and waste infrastructure in India involves building and maintaining public and private toilets, sewage systems, and wastewater treatment plants to ensure proper disposal of human waste, prevent environmental pollution, and promote public health in urban areas. As the coverage of toilets has grown rapidly owing to SBM 1.0 and 2.0, government focus has shifted to sustaining the use of the created infrastructure.

Key urban toilet infrastructure statistics

13%

Share of households that do not use a common facility in the building

3%

Share of households that do not have access to toilets

3%

Share of households that use a community facility

81%

Share of households that use an improved, not shared facility

Source: MoSPI, Multiple Indicator Survey of India 2020-21, 2023

The government has exceeded toilet construction targeted in urban areas:

By 2023, the SBM-U scheme had constructed nearly 63 lakh individual household latrines and about 600,000 community toilets (CTs) as against the respective targets of 59 lakh individual latrines and 500,000 CTs.⁶ As a result, the proportion of urban households without access to toilets dropped from 18.6% in the 2011 census to only 2.9% in 2020-21.⁷

Household urban toilet construction has seen growth:

Urban residents with access to improved toilet facilities that are non-shared increased from 70% in 2014-15 to 81% in 2019-21.⁸

Urban Handwashing with Soap

Urban handwashing with soap in India focuses on promoting and providing facilities for proper hand hygiene in urban settings. It includes the establishment of handwashing stations, awareness campaigns,

³ MoSPI, Multiple Indicator Survey in India, 78th Round, (2020-21), 2023

⁴ Sufficient water in urban areas is 135 litres per capita per day (lpcd). In rural areas it is 55 lpcd. Details here: Ministry of Jal Shakti, Per capital availability of water, 2020

⁵ MoSPI, Multiple Indicator Survey in India, 78th Round, (2020-21), 2023; MoSPI, Drinking Water, Sanitation, Hygiene and Housing Condition in India, NSS 76th Round, (July 2018-December 2018), 2019

⁶ Swachh Bharat Mission Urban, Achievements, accessed June 2023

⁷ MoSPI, Multiple Indicator Survey of India 2020-21, 2023; ORGI, Availability and type of latrine facility -2001, 2011, 2022

⁸ Institute for Population Sciences (IIPS) and ICF, National Family Health Survey (NFHS-5), 2019-21: India: Volume I, 2022

and education to encourage the habit of regular handwashing with soap. Focused efforts to promote handwashing with soap through targeted programs, behaviour change, and infrastructure upgrades can significantly improve hygiene practices and health in vulnerable urban communities.

Key urban hand washing statistics



Source: CBGA, IRC and UNICEF, Commitments towards hand hygiene in India: A National level analysis, 2022; Ministry of Jal Shakti, National Annual Rural Sanitation Survey (NARSS), 2019-2020; Ministry of Statistics & Programme Implementation, Drinking Water, Sanitation, Hygiene and Housing Condition in India, 2018

Several government programs have emphasized handwashing with soap:

Several are school-focused, such as the Swachh Bharat Swachh Vidyalaya campaign, promoting hand hygiene among children.⁹ The National Urban Health Mission also integrates hygiene education and infrastructure for handwashing in schools, Anganwadis, and community spaces.¹⁰

Household access to handwashing facilities with water and soap has improved in urban areas:

As per National Sample Survey 2020-2021, ~93% of people in urban areas reported having access to handwashing facilities with water and soap or detergent within their premises.¹¹ This is a marked improvement from the 80% who had access to basic handwashing facilities including soap and water in urban India in 2017.¹²

Urban Menstrual Hygiene Management

Menstrual hygiene management (MHM) refers to access to safe menstrual materials, accurate information, and facilities for washing, waste management, and bathing during menstruation. In urban India, access to menstrual hygiene products and facilities is better than in rural areas. However, challenges related to awareness, and quality of services impede safe MHM for poor and marginalized girls and women.

Key menstrual hygiene management statistics



Source: UNICEF, Changing the future with lessons from the past, accessed August 2023; UNESCO, UNESCO Joins Whisper®'s mission to #KeepGirlsInSchool to ensure girls don't miss out on a promising future, 2021; Ministry of Health and Family Welfare, National Family Health Survey (NFHS-5), 2021

Numerous initiatives have been implemented to enhance the accessibility and affordability of menstrual hygiene products across the nation:

To improve the supply of affordable menstrual products, the Pradhan Mantri Bhartiya Janaushadhi Pariyojana (PMBJP) program introduced Jan Aushadhi Oxo-biodegradable Sanitary Napkins costing only Rs 1 per pad, in over 9,400 PMBJP Kendras across the country. In 2018, an important step was taken to exempt sanitary napkins from Goods and Services Tax.¹³ This move is a significant improvement in promoting menstrual hygiene and making sanitary napkins more affordable and accessible.¹⁴

⁹ Ministry of Education, SoP for sustaining WASH in schools, 2021; Ministry of Human Resource Development, Swachh Bharat Swachh Vidyalaya, 2014

¹⁰ Ministry of Health & Family Welfare, National Health Mission, accessed June 2023

¹¹ Ministry of Statistics & Programme Implementation, PIB, Multiple Indicator Survey (MIS) - NSS 78th Round Report [2020-21], 2023

¹² World Bank, Many homes lack basic handwashing facilities, 2020

¹³ Press Information Bureau, 2018

¹⁴ Press Information Bureau, 2018

The use of hygienic menstrual protection methods has improved in urban India:

As per the National Family Health Survey 2019-2021, ~89% of women in the age group of 15-24 years in urban India were using locally prepared napkins, sanitary napkins, tampons, or menstrual cups during menstruation, as compared to ~77% women in the same age group in 2015-2016.¹⁵

Use of cloth as protection during menstruation persists in urban areas:

32% of women in urban India, aged 15-24, opt for cloth as their menstrual protection method.¹⁶ This is a noticeable decrease compared to 2015-16, where ~43% of urban women in the same age group relied on cloth for menstrual hygiene.¹⁷ *(Note: respondents were allowed to report multiple methods of menstrual protection in the survey)*

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¹⁵ Ministry of Health & Family Welfare, National Family Health Survey 5.0, 2019-2021; Ministry of Health & Family Welfare, National Family Health Survey 4.0, 2015-2016

¹⁶ Ministry of Health & Family Welfare, National Family Health Survey 5.0, 2019-2021; Ministry of Health & Family Welfare, National Family Health Survey 4.0, 2015-2016 (Note: respondents were allowed to report multiple methods of menstrual protection and hence, there might be some overlap with those using non-hygienic methods of protection)

¹⁷ Ministry of Health & Family Welfare, National Family Health Survey 5.0, 2019-2021; Ministry of Health & Family Welfare, National Family Health Survey 4.0, 2015-2016

Sub-sectoral trends: Rural WASH

This section presents a summary view of relevant statistics and trends related to the following sub-sectors within the urban WASH sector:

- Rural ODF waste management
- Rural tap water supply
- Rural ODF and toilet infrastructure
- Rural handwashing with soap
- Rural menstrual hygiene management

Rural ODF+ and Waste Management

Rural waste management in India involves the organized collection, disposal, and treatment of waste in through decentralised solutions. Efforts focus on waste segregation, promoting sustainable waste practices, and creating awareness about the environmental impact of waste. SBM-G 2.0 and the NFSSM have and continue to support the implementation of nationwide waste management in rural India.

Key ODF + Rural Waste Management statistics

5.08 lakh+
Villages declared ODF Plus*

96,000+
ODF + Model** Villages

Source: Ministry of Jal Shakti, [SBM-G Dashboard](#), accessed December 2023

Note: * An ODF Plus village is one which has sustained its Open Defecation Free (ODF) status along with implementing either solid or liquid waste management systems.

**An ODF Plus Model village is one which is sustaining its ODF status and has arrangements for both Solid Waste Management and Liquid Waste Management; and observes visual cleanliness

Growing fraction of villages have arrangements for waste management:

As of January 2024, 47% of villages had arrangements for SWM, while 75% had arrangements for LWM.¹⁸

Annual progress for development of waste management infrastructure has been slower compared to targets:

The Standing Committee on Water Resources noted that in 2021-2022, only 22% of villages (46,347 out of the targeted 2,07,945) were covered by SWM infrastructure, falling short of the targets set under the SBM-G.¹⁹

Rural Tap Water Supply

Rural tap water supply focuses on the provision of accessible and clean drinking water to rural households via functional taps, pipelines, and water extraction and treatment infrastructure. The Jal Jeevan Mission (JJM) has made significant progress towards improving availability of tap water connections in rural areas.²⁰

Key rural tap water supply statistics

63%
Rural households using a functional household tap connection (FHTC)

86%
Working tap connections in rural households*

74%
Rural households receiving water on a daily basis

87%
Water received by rural households in potable

Source: Department of Drinking Water and Sanitation, Functionality Assessment of Household Tap Connections, 2022

Note: * Working tap connections involve those households which received water through FHTC at least once in the last 7 days

¹⁸ Department of Drinking Water and Sanitation, Swachh Bharat Mission (Grameen) 2.0, accessed Jan 2024

¹⁹ Lok Sabha Secretariat, Standing Committee on Water Resources, 2023

²⁰ Jal Jeevan Mission, Functional Household Tap Connection (FHTC) in every rural home, accessed 17th June 2023

Tap water connections have increased since the launch of JJM:

Only ~17% of rural households in India (32 million) had tap water connections in August 2019, when JJM was launched.²¹ Under the scheme, by June 2023, approximately 63% of households (123 million) across India had tap water connections.²² In 2020, a 100-day campaign was launched to provide tap water supply in schools and AWCs.²³ In less than ten months, in India, tap water connections had been made in 66% of schools, 60% of AWCs, and 69% of GPs and CHCs.²⁴ As of June 2023, the number stood at ~89% schools, ~85% AWCs, and ~77% GPs/ CHCs.²⁵

Some states need to accelerate coverage of tap water connections:

The Standing Committee on Water Resources, while noting that many states had reached 90-100% coverage of rural households with tap water connections, expressed concerns that states like Uttar Pradesh, Chhattisgarh, Jharkhand, West Bengal, Kerala, and Tamil Nadu, had provided less than 40% of households with tap connections by March 2022.²⁶ As of June 2023 too, some states such as West Bengal, Jharkhand, and Lakshadweep had less than 40% households with such connections.²⁷

Rural ODF and Toilet Infrastructure

Rural ODF and Toilet Infrastructure development focuses on providing households with access to toilets and improving sanitation facilities in rural areas. The focus has now shifted to helping villages achieve ODF Plus (+) status by ensuring continued access to sanitation facilities and safe disposal of liquid and solid waste, as part of the second phase of SBM 2.0

Key Rural ODF and Toilet Infrastructure statistics

~69%

Rural households have non-shared access to toilets

~9%

Rural households have access to shared toilet facilities

5.08 lakhs

ODF Plus villages in India

Source: MoSPI, Multiple Indicator Survey of India 2020-21, 2023; Ministry of Jal Shakti, India Achieves Major Sanitation Milestone As 75% Villages Are Now ODF Plus under Swachh Bharat Mission - Gramin, 2023, Ministry of Jal Shakti, SBM Dashboard, accessed December 2023

Note: An ODF Plus village is one which has sustained its Open Defecation Free (ODF) status along with implementing either solid or liquid waste management systems.

A rapid increase in household toilets constructed:

In 2014, about 10 crore rural and 1 crore urban households lacked access to sanitary toilets. By June 2023, under SBM-G 2.0, nearly 11.2 crore household toilets and over 2.3 lakh community sanitary complexes have been constructed in rural India.²⁸

Utilisation rate of toilets by households is high:

As per the 76th edition of the National Sample Survey, among households having access to latrine, ~95% males and ~96% females in rural areas used latrines regularly, reflecting negligible gender gap in use.²⁹ As per the National Annual Rural Sanitation Survey 2019-2020, ~90% of people who had access to toilets were using toilets always.³⁰

Rural Handwashing with Soap

Rural handwashing in India focuses on promoting and facilitating proper hand hygiene practices in rural areas by building awareness, providing access to handwashing facilities, and encouraging regular

²¹ Jal Jeevan Mission, Functional Household Tap Connection (FHTC) in every rural home, accessed 17th June 2023

²² Jal Jeevan Mission, Functional Household Tap Connection (FHTC) in every rural home, accessed 17th June 2023

²³ Ministry of Jal Shakti, Tap Water Supply Reaches 66% Schools & 60% Anganwadi Centres, 2021

²⁴ Ministry of Jal Shakti, Tap Water Supply Reaches 66% Schools & 60% Anganwadi Centres, 2021

²⁵ Jal Jeevan Mission, Tap connection in schools/ AWCs/ GPs/ CHCs, accessed 17th June 2023

²⁶ Lok Sabha Secretariat, Standing Committee on Water Resources, 2023

²⁷ Jal Jeevan Mission, Functional Household Tap Connection (FHTC) in every rural home, accessed 17th June 2023

²⁸ Ministry of Jal Shakti, Swachh Bharat Mission (Gramin) 2.0, 2023

²⁹ Ministry of Statistics & Programme Implementation, National Sample Survey 76th Round, 2018

³⁰ Ministry of Jal Shakti, National Annual Rural Sanitation Survey, 2019-2020

handwashing with soap. The SBM-G mission and the Swachh Aadat, Swachh Bharat campaign have significantly enhanced handwashing with soap after use of toilet.

Key rural handwashing statistics

89.8%

Rural residences with handwashing facilities with water in 2019-21

68.2%

Rural residences with handwashing facilities with soap in 2019-21*

98.7%

Rural schools with observed evidence of handwashing in 2019-20

36%

Rural schools where water and soap are both available near handwashing facilities in 2019-20

Source: Ministry of Health & Family Welfare, National Family Health Survey (NFHS-5), 2022; Ministry of Jal Shakti, National Annual Rural Sanitation Survey, 2019-2020

Note: * Soap includes soap or detergent in bar, liquid, powder or paste form.

Over 3/4th of households in rural India have access to handwashing facilities with water and soap:

As per National Sample Survey 2020-2021, ~77% of people in rural areas reported having access to handwashing facilities with water and soap/detergent within their premises.³¹

Uptake of handwashing in rural schools is high:

Evidence of handwashing was observed in ~99% percent of rural schools in 2019-2020 overall.³²

Rural Menstrual Hygiene

Menstrual hygiene management (MHM) refers to access to safe menstrual materials, accurate information, and facilities for washing, waste management, and bathing during menstruation.³³ Unsafe management of menstruation has been a significant challenge in rural India. While the government provides subsidized pads across rural India, campaigns for awareness generation, better products/services, upgraded infrastructure, and supportive policies can further support rural MHM.

Key rural menstrual hygiene statistics

72.3%

Share of rural women (15-24 years) that use hygienic methods of protection during their menstrual period in 2019-21*

96%

Share of rural women (15-19 and 20-24) take a bath during their menstrual period

47 crore

Jan Aushadhi Suvidha Sanitary Pads sold under Pradhan Mantri Bhartiya Janaushadhi Pariyojana during 2018-23

Source: Ministry of Health & Family Welfare, National Family Health Survey (NFHS-5), 2019-2021; Ministry of Chemicals and Fertilizers, Suvidha Sanitary Napkins under PMBJP, 2023

Note: * Locally prepared napkins, sanitary napkins, tampons, and menstrual cups are considered to be hygienic methods of protection.

Multiple efforts have been undertaken to make menstrual hygiene products accessible and affordable in the country:

The Department of Pharmaceuticals, through Pradhan Mantri Bhartiya Janaushadhi Pariyojana (PMBJP) Kendras has been providing Jan Aushadhi Oxo-biodegradable Sanitary Napkins at Rs 1/ per pad across 9,400 PMBJP Kendras in the country.³⁴ Similarly, under the Scheme for Promotion of Menstrual Hygiene, a pack of sanitary napkins is provided to adolescent girls by the ASHA workers at subsidized rate of Rs. 6 per pack.³⁵ States such as Maharashtra and Chhattisgarh have also implemented decentralized models wherein SHG federations and clusters procure pads from vendors, and SHG women actively distribute sanitary pads at the community level.³⁶ The Operational Guidelines for Promotion of Menstrual Hygiene among Adolescent Girls (10-19 Years) in Rural Areas by the National Rural Health Mission also states that

³¹ Ministry of Statistics & Programme Implementation, PIB, Multiple Indicator Survey (MIS) - NSS 78th Round Report [2020-21], 2023

³² Ministry of Jal Shakti, National Annual Rural Sanitation Survey, 2019-2020

³³ UNICEF, Guidance on Menstrual Health and Hygiene, 2019

³⁴ Department of Pharmaceuticals, Jan Aushadhi Suvidha Napkins, accessed June 2023

³⁵ Ministry of Women and Child Development, PIB, Scheme for Promotion of Menstrual Hygiene, 2022

³⁶ Government of Maharashtra, ASMITA YOJANA, accessed June 2023; Water Aid, Strengthening the Supply Chain of Menstrual Hygiene Products in India, 2021

existing mechanisms of the School Health Programme may be used when sanitary napkins are to be distributed through the school route.³⁷

Use of hygienic methods of protection during menstrual period has improved in rural India:

As per the National Family Health Survey 2019-2021, ~73% of women in the age group of 15-24 years in rural India were using locally prepared napkins, sanitary napkins, tampons, or menstrual cups during menstruation, as compared to only ~48% women in the same age group in 2015-2016.³⁸

Cloth continues to be widely used for protection during menstruation:

As per the National Family Health Survey 2019-2021, 57% of women in rural India in the age group of 15-24 years reported using cloth as protection during the menstruation period.³⁹ Comparatively, in urban areas, 32% of women reported cloth during the same period.⁴⁰

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³⁷ National Health Mission, Promotion of Menstrual Hygiene among Adolescent Girls (10-19 Years) in Rural Areas, 2010

³⁸ Ministry of Health & Family Welfare, National Family Health Survey 5.0, 2019-2021; Ministry of Health & Family Welfare, National Family Health Survey 4.0, 2015-2016 (Note: respondents were allowed to report multiple methods of menstrual protection and hence, there might be some overlap with those using non-hygienic methods of protection)

³⁹ Ministry of Health & Family Welfare, National Family Health Survey 5.0, 2019-2021; Ministry of Health & Family Welfare, National Family Health Survey 4.0, 2015-2016 (Note: respondents were allowed to report multiple methods of menstrual protection and hence, there might be some overlap with those using non-hygienic methods of protection)

⁴⁰ Ministry of Health & Family Welfare, National Family Health Survey 5.0, 2019-2021; Ministry of Health & Family Welfare, National Family Health Survey 4.0, 2015-2016 (Note: respondents were allowed to report multiple methods of menstrual protection and hence, there might be some overlap with those using non-hygienic methods of protection)

Opportunities and challenges

Opportunities

Urban WASH

 Improvement in health outcomes	• Inculcating the practice of handwashing in school settings can help break disease transmission (by washing hands before eating mid-day meals) and can imprint good hygiene behaviour in children • Improving MHM knowledge amongst women and girls and increasing access to safe and affordable sanitary materials improves overall reproductive and sexual health, including maternal outcomes, lower teen pregnancy, and improved fertility • Improving access to proper sanitation facilities like toilets, encourages hygienic practices and improves public health • Ensuring potable water supply in all households can drive health outcomes and lower health expenditure
 Increase in economic outcomes and workforce productivity	• Investing in workplace sanitation services increases employee health, leading to fewer sick days and improved productivity • Improved access to piped water reduces women's time burden and drudgery, thus alleviating their poverty and increasing participation other productive market activities • Instituting inclusive menstrual facilities and policies improve company retention and performance and foster employee appreciation and commitment • Improving infrastructure for sewage treatment can help reduce economic losses caused by inadequate sanitation, equivalent to 6.4% of India's GDP⁴¹ • Mechanising cleaning and providing safety measures for manual scavengers improves their livelihoods and living standards, thus enhancing economic outcomes for these sanitation workers
 Improvement in social equity and education outcomes	• Presence of separate toilets for females in schools alongside other menstrual hygiene facilities can help reduce dropout and absenteeism • Reducing women's time and labour needed for water collection by better access to piped water can enable their participation in education • Implementing school-level interventions through adolescent health education and sensitization programs covering menstruation and puberty can improve girls' knowledge and help tackle stigma
 Improvement in climate resilience	• Maximising resource recovery from waste, both material and energy, along with proper disposal, ensures effective waste management and contributes to significant GHG emission reductions

⁴¹ WEF, India's building more toilets, but what happens 'after the flush?', 2019

	• Minimising water losses through leak detection and better infrastructure can help meet increasing water demands, ensure equitable water distribution, and save lives
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Rural WASH

 Improvement in health outcomes	• Providing hygienic sanitation facilities and promoting regular handwashing practices in rural areas can reduce disease transmission, thereby enhancing health outcomes • Using effective waste management practices can significantly reduce disease occurrence, leading to improved public health • Ensuring safely managed drinking water for all Indian households through the 'Har Ghar Jal' program could avert ~400,000 deaths from diarrheal diseases and prevent ~14 MN Disability Adjusted Life Years related to such diseases • Improving access to safe and affordable sanitary materials can reduce risk of infections, improve fertility, lower teen pregnancy, and improve overall reproductive and sexual health
 Greater economic productivity and outcomes	• Improving sanitation facilities results in fewer disease cases and time optimisation (like due to a closer toilet), resulting in fewer medical expenses and increased daily efficiency in activities • Leveraging efficient SLWM through reusing/recycling products can result in financial benefits • Expanding tap water connections to all rural households can positively change number of employment days of adult members and impact income, thereby improving economic productivity
 Improvement in social equity and awareness	• Improving toilet infrastructure provides women with appropriate sanitation options, fostering dignity, security, and active participation in society • Improving drinking water access in households through taps can help reduce responsibilities like fetching water for women, and associated care burdens • Initiating community outreach and awareness generation efforts for women and girls through health workers and SHGs can promote greater MHM knowledge and safe practices at the household level • Initiating MHM-related awareness generation amongst men can unlock strong psycho-social support for menstruating women and girls, thus creating an enabling environment to address needs
 Higher education outcomes	• Improving toilet facilities in educational institutions can enhance student attendance, learning outcomes, and education standards • Installing child-friendly handwashing stations near school toilets, hosting handwashing sessions, and integrating hygiene education can spread awareness and instil positive habits in children, leading to greater health and better school performance • Improving provision of sanitary pads, private changing rooms and related WASH infrastructure in schools, greater awareness-generation for adolescent girls and boys alongside teacher training to provide menstrual counselling, can facilitate safe MHM practices, break taboos and improve retention of girls in schools

Challenges

Urban WASH

 Limited availability, access, and quality of facilities and related infrastructure	• Insufficient facilities like toilets, running water, and proper containment systems in urban households lead to persistence of open defecation • High maintenance costs of existing toilet structures (like OSS systems) dissuade poor and low-income households from using toilets, leading to inequitable gaps in usage • Poor location, inappropriate design, and inadequate maintenance of community and public toilets lead to challenges faced by women and adolescent girls • Inadequacy in industrial solid waste disposal due to lack of requisite infrastructure leads to inefficient wastewater management • Insufficiently maintained last-mile pipelines that are often laid near stormwater drains or municipal sewers lead to quality deterioration of water supplied from municipal plants • Lack of handwashing facilities by urban population or lack of either soap or water in existing facilities affect personal hygiene • Lack of affordable menstrual products or gaps in supply alongside poverty restricts urban women from using hygienic methods of protection during menstruation • Restricted access to safe and private spaces in urban slums and apt disposal mechanisms also exposes women and adolescent girls to specific vulnerabilities concerning menstruation
 Institutional and ecosystem-related barriers	• The largely unregulated nature of India's e-waste collection, transportation, processing, and recycling value chain due to domination by the informal sector leads to inefficient practices • Lack of demand to treat greywater and high capital requirements for set up and R&D have led to limited enterprise presence • Heavy dependence on older technologies for handling waste management due to limited funds push adoption of technologies with lower capital costs despite poor performance parameters • Inadequacy of trained professionals in the sanitation delivery ecosystem like certified plumbers, mechanical engineers, etc. affects efficiency and service delivery • Inefficient water pricing mechanisms and high NRW levels affect India's water security • Lack of standardised data and public access on key indicators of water supply leads to challenges in monitoring, comparison, or evidence-based decision-making
 Limited awareness and restrictive social norms	• Lack of awareness about hygiene, water conservation, and safe sanitation practices underscores the need for robust awareness-generation initiatives regarding the right practices

	• Existence of social prejudice against sanitation workers leads to discrimination and limited access to their basic rights/services, hindering their overall well-being • Limited awareness amongst citizens around waste management results in persistence of practices like dumping of waste and open burning in cities, towns and communities, resulting in contamination of water sources, air pollution, and soil degradation • Limited awareness about importance of handwashing with soap affects hygiene practices of people in urban India • Limited menstruation-related awareness amongst women and girls, as well as men hinders adoption of safe MHM practices and creation of a support system for them • Restrictive socio-cultural norms around menstruation contribute to the social isolation of women and adolescent girls, while awareness gaps and misconceptions prevent open discussions around safe menstrual practices by parents, teachers, and other community members
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Rural WASH

 Limited availability, access, and quality of facilities and related infrastructure	• Insufficiency in number of toilets in schools affects student hygiene practices • Limited access to hand washing facilities with water and soap/detergent within premises affects sanitation in rural areas • Limited availability of water and soap for handwashing in schools in rural India affect student health and hygiene practices
 Institutional and ecosystem-related barriers	• Inadequate functional waste collection and disposal systems at the Panchayat level in India affect rural sanitation and health • Inefficiencies faced by states in establishing tap connections include procedural challenges such as delays in obtaining clearances for project execution and inability of states to provide a matching share in funds, thus hindering related implementation • Inadequacies in setting up and functioning of bodies related to village water supply, like GPs or their sub-committees such as Village Water And Sanitation Committee (VWSC) and Pani Samiti as under the JJM, hinder planning, implementation, management, O&M of in-village water supply • Insufficient number of Community Water Purification Plants (CWPP) set up by states/UTs under JJM in habitations where water is adversely impacted by contaminants such as arsenic and fluoride, adversely affect water quality • Low adherence to waste 'segregation at source' practices affect efficient waste disposal- over half of all households nationwide reported having disposed of the waste outside a common system⁴² • Lack of use of mechanised equipment by desludging operators and trucks lead to improper waste handling, posing health risks to operators and handlers

⁴² Ministry of Jal Shakti, NARSS, 2019-20

	• Lack of transportation of sludge and septage to a Sewage Treatment Plant or FSTP results in practices like frequent illegal dumping on land grounds and water bodies
 Limited awareness and restrictive social norms	• Restrictive societal attitudes and taboos limit uptake and usage of toilets in rural India, thereby leading to persistence of open defecation even in villages with adequate toilet access • Lack of sustainability of handwashing practices observed as these sound practices increase during epidemics, but tend to wane with time and are usually not sustained, thus highlighting the knowledge-action gap

Institutional framework and policy landscape –

Key ministries and departments

Water falls under the State List of the Constitution, giving legislative assemblies the power to make laws on the subject within the prescribed guidelines of the centre.

The central government is responsible for formulation of broad policies and framework that guide states in managing their water resources, resolving international and inter-state water conflicts, providing financial assistance to states, and monitoring and regulating water resources in India. At the centre, the Ministry of Jal Shakti is the nodal ministry responsible for the overall planning and policy formulation related to water resources in India. It aims to ensure sustainable and equitable management of water, both in terms of availability and access. It is further bifurcated into two departments with distinct mandates concerning the different uses of and concerns over water resources.

The Department of Water Resources, River Development and Ganga Rejuvenation (DoWR, RD & GR) is the nodal agency for water resource management, responsible for building national planning, perspective, policies and programmes for the development, conservation, and management of water as a national resource. The Department of Drinking Water and Sanitation (DoDWS) provides technical and financial assistance to the States to provide safe and adequate drinking water to rural India with focus on service delivery.

Central government institutions such as the Central Water Commission and the CPCB are also involved in different aspects of water resource management such as mobilizing and facilitating scientific, and technical inputs and finance across states, regulating water quality, reviewing national policies from time to time, responding to the contemporary challenges to water resources, conducting research and development, and monitoring data for forecasting.

The central government also addresses inter-state water sharing disputes by establishing tribunals under the ISRWD Act of 1956. Currently, there are five active tribunals. After the resolution of disputes by the tribunal, suitable mechanisms like control authorities, supervisory committees, or boards are established to ensure the implementation of the award. These regulatory organisations such as Godavari River Management Board, Upper Yamuna River Board function under the DoWR, RD & GR.

At the state level, the respective governments legislate and implement policies based on broad guidelines provided by national frameworks such as National Water Policy and Model Bills.

Multiple state ministries and departments are involved in water management at state level as well. Departments of various ministries such as Department of Water Resources, Department of Agriculture/Horticulture, Department of Rural Development and State Disaster Management Authorities that lead efforts on policy and program implementation on groundwater management, irrigation water management. State level institutions like the State Water Boards, Public Works Department, and the State Pollution Control Board conduct functions such as water quality monitoring, and monitoring of data, preparations and response for hydro-met natural disasters.

Key ministries and departments: Urban WASH



Ministry of Housing and Urban Affairs

Oversees and manages the supply of drinking water, rejuvenation of water bodies, waste management, and rainwater harvesting in urban areas.



Ministry of Environment, Forests, and Climate Change

Promotes the safe management and use of hazardous substances and waste; sets and notifies rules and guidelines for waste management and provides financial assistance to states to run awareness programmes.



Ministry of Health & Family Affairs

Guides and implements measures on sanitation and hygiene, including menstrual health management.



Ministry of Education

Ensures well-functioning water, sanitation, and hygiene facilities in schools and raises sanitation and hygiene awareness.



Central Public Health and Environmental Engineering Organisation

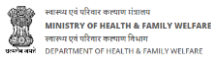
Provides technical inputs in policy formulation to the MoHUA on matters related to urban water supply and sanitation, including solid waste management, and supports the states by providing technical advice, guidelines and propagating new technologies in the field.

Key ministries and departments: Rural WASH



Department of Drinking and Sanitation, Ministry of Jal Shakti

Oversees efforts to provide adequate, regular, and long-term drinking water supply to rural households and achieve universal sanitation cover in rural areas of the country; guides and supports state initiatives on menstrual health management.



Ministry of Health & Family Affairs

Guides and implements measures on sanitation and hygiene, including menstrual health management.



Ministry of Education

Ensures well-functioning water, sanitation, and hygiene facilities in schools and raises sanitation and hygiene awareness.



Ministry of Panchayati Raj

Builds capacity of Panchayati Raj Institutions (PRIs) to deliver WASH services to the community.



Central Pollution Control Board

Provides technical assistance to states for the prevention, control, and abatement of water pollution in accordance with the Water (Prevention and Control of Pollution) Act, 1974.

Major policies and programs: Urban WASH

This section highlights existing policies related to the urban WASH sector and their most prominent features. A more comprehensive list of policies is provided in the policy repository of the NITI Knowledge Platform.



Swachh Bharat Mission- Urban⁴³

SBM-U targets effective solid waste management, sustainable sanitation, and wastewater treatment. It also aims to create awareness at a large scale and institutional capacity to deliver the programmatic interventions under this scheme.



Atal Mission for Rejuvenation and Urban Transformation 2.0 (AMRUT 2.0)⁴⁴

AMRUT strives for universal coverage of functional household water supply taps in all the statutory towns of the country and sewerage/septage coverage for the 500 cities covered in its first phase. It focuses on:

- Promoting the circular economy of water through City Water Balance Plans (CWBP), which focus on the reuse and recycling of treated sewage, rejuvenation of water bodies, and conservation of water
- Reforming agenda for improving citizens' quality of life, reducing non-revenue water, and strengthening urban finance



National Urban Livelihoods Mission (NULM)⁴⁵

It aims to facilitate access to drinking water and sanitation services for the urban homeless and develop informal vendors' markets with water supply and waste management facilities.

Major policies and programs: Rural WASH



Jal Jeevan Mission (JJM)⁴⁶

Ministry of Jal Shakti

The JJM aims to supply regular, adequate drinking water for the long term to every rural household in the country. The program also aims to enhance source sustainability measures such as recharge and reuse through greywater management and rainwater harvesting based on a community approach to water and the use of information, education, and communication.



Swachh Bharat Mission-Grameen Phase II (SBM-G)⁴⁷

Ministry of Jal Shakti

SBM-G aims to extend the impact of the first phase of SBM to sustain the ODF behaviour and provide interventions for SLWM in villages.



National Rural Health Mission (NRHM)⁴⁸

Ministry of Health and Family Welfare

The NRHM aims for universal access to public services encompassing sanitation and hygiene to ensure the effective integration of health concerns in rural areas of the country.

⁴³ MoHUA, Swachh Bharat Mission - Urban Guidelines, 2021

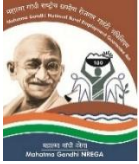
⁴⁴ MoHUA, AMRUT 2.0: Operational Guidelines, 2021

⁴⁵ MoHUA, National Urban Livelihoods Mission: Mission Document, accessed June 2023

⁴⁶ Ministry of Jal Shakti, Jal Jeevan Mission, accessed July 2023

⁴⁷ Department of Drinking Water & Sanitation, Swachh Bharat Mission - Grameen, accessed July 2023

⁴⁸ Ministry of Health & Family Welfare, National Health Mission, accessed July 2023



Mahatma Gandhi National Rural Employment Guarantee Act (MGNREGA)⁴⁹

Ministry of Rural Development

The MGNREGA works cover rural sanitation projects, including toilet building, soak pits, and solid and liquid waste management.

DRAFT

⁴⁹ Ministry of Law & Justice, Mahatma Gandhi National Rural Employment Guarantee Act, 2005

Way forward

India has made remarkable progress on increasing WASH access in both urban and rural areas over the past decade. However, as highlighted in the earlier sections, gaps persist across rural and urban areas to achieve sustainable universal coverage, address social inequities and entrench positive habits. Some key areas for intervention going forward include:

 Increase investments in decentralized waste management	Greater investments should be channeled towards strengthening decentralized solid and liquid waste management models through blended financing mechanisms including commercial funds, donor grants and government incentives. This can shore up infrastructure, ensure financial viability of projects and drive scientific waste disposal.
 Enhance behaviour change programming through behavioural science	Behaviour change efforts need enhancement through sustained, hyperlocal campaigns that are anchored in behaviour science, to transform social attitudes on toilet usage, handwashing practices, waste segregation and safe menstrual hygiene.
 Accelerate private sector participation in WASH	Enabling policy frameworks must be instituted to promote private sector participation across the WASH delivery chain – right from construction to operations and maintenance. Simultaneously, capacities of rural and urban institutions require strengthening for better planning, implementation and monitoring at the grassroots.
 Increased use of digital tools for service delivery	Leveraging emerging digital tools around remote sensing, Internet-of-Things and mobile applications can support real-time transparent data capture on service delivery alongside predictive analytics for preemptive action. However, technology integration should still keep the user at the center through vernacular interfaces.
 Enhanced menstrual hygiene management practices	As a theme, menstrual hygiene management needs dedicated attention in both urban and rural areas. Behaviour change focused on MHM will be vital to shift norms and habits. Business models that bring safe MHM products should be encouraged across the country. Finally, funding models that increase access to these products should be prioritized to ensure that no girl or woman gets left behind.

Recommended readings

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